One-pot PCR amplicon library workflow

gene-region primers + barcode primers in the same tube → pool → Nanopore (Evoprep) · HuangLab LIMS stages 1 → 3a → 3b → 4a

Stage 1 — Specimen# / Culture#

Specimen registered in LIMS
soil / environmental → Specimen# | fungal strain → Culture#

DNA extraction

Stage 3a — DNA extraction · MolWork#

DNA template → MolWork#
soil crude extract (SuperSoil) | pure fungal extraction (BOMB v9)
MolWork# minted after quantification

Soil DNA extraction
SuperSoil protocol

Fungal DNA extraction
modified BOMB v9.0

Set up one-pot PCR

One reaction tube (per sample)

gene-region primers
(head-tailed, 0.1 µM each)

+

barcode primers
(barcode + head seq, 0.3 µM each)

50 µL: KAPA HiFi 2X + both primer sets + 3 µL template

Stage 3b — PCR + gel + Qubit · pcr_reaction row

One-pot PCR
phase 1: 25 cycles, anneal 60–66 °C (gene primers)
phase 2: 15 cycles, anneal 65 °C (barcode primers)

One-pot PCR protocol
50 µL mix + 25/15-cycle program

One-pot PCR products: one tube per sample, one barcode per tube

Tube 1 barcode 1 · **Tube 2** barcode 2 · **Tube 3** barcode 3

barcode — head seq — gene — head seq — barcode

Gel check
confirm barcoded product

Qubit Flex
quantify → 10 ng/µL (Stage 3b)

LIMS register
`/lims-register`
Stage 3b PCR row → second_pcr_ok + barcode

conc. → equal-molar volumes

Pooling (equal molar)

Pool all barcoded samples
equal molar of every sample

Pooled sample

Stage 4a — Nanopore experiment · Nanopore#

Purification (bead clean-up)
PCR sample : bead = 1 : 1

Size selection 1X
bead purification protocol

Purified pooled sample

Ready to be sequenced
library construction using Evoprep

EvoPrep library prep
KAPA EvoPrep NBD114.96

Legend

- Head seq (universal tail on the gene-region primers)
- Barcode (sample-specific, on the barcode primers)
- Gene region (target amplicon; colour = different sample)

QR codes (left) open the lab's self-hosted protocol PDF for that step
left bands = HuangLab LIMS stages (/lims-register); dashed = QC / info flow
single gel check: no intermediate 1st-PCR product to inspect